# SENA: Search Engine with Neural Attention for Long COVID Retrieval

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**Project Proposal of STATS 507 Course with Dr. Xian Zhang**

## I. Overview:

Since the first reported coronavirus disease 2019 (COVID-19) patient in December 2019 in China [1], over 778 million cases and 7.1 million deaths have been recorded worldwide [2]. Though the majority of COVID-19 patients experience mild symptoms, about 20% of cases have severe negative outcomes involving hospitalization, particularly those with underlying comorbidities such as diabetes and cardiovascular disease [3].

Beyond the acute phase of infection, a clinical concern is the condition known as *long COVID*, which refers to persistent, new, or relapsing symptoms that continue for weeks or months after initial recovery from COVID-19. Despite increasing recognition, defining and diagnosis of long COVID remains challenging. There is no universally accepted time threshold for diagnosis, recovery timelines vary widely across patients, and many infections were never clinically confirmed due to asymptomatic cases or lack of testing [4]. Furthermore, the absence of standardized terminology and the continuous emergence of new findings make it difficult to systematically identify relevant long COVID literature using traditional keyword-based search methods [5]. This challenge is amplified by the rapid expansion of COVID-19 research. According to the Dimensions database [6], more than three million COVID-19-related studies have been published to date, including over 200,000 journal articles and preprints by the end of 2020 alone [7]. Identifying long COVID studies within millions of COVID-related publications, many of which are preprints with limited or inconsistent keyword annotations, is essential for improving our understanding of long COVID.

Motivated by these challenges, this project aims to leverage artificial intelligence (**AI**) in text mining, particularly natural language processing (**NLP**) and deep learning (**DL**), to improve the identification of long COVID-related research articles. The goal is to develop an AI-based classification approach that can distinguish long COVID studies from non-long COVID literature using article titles and abstracts as input. This approach is expected to outperform traditional keyword-based search methods by capturing deeper semantic patterns in the text. For this purpose, we utilize a labeled dataset available on Hugging Face [8], originally published by Langnickel et al. [5], which contains 552 COVID-19-related articles annotated for long COVID relevance. Each data instance consists of an abstract and a binary label indicating whether the study is related to long COVID. By applying ML and NLP techniques, we aim to build a robust search and classification tool that enables more efficient retrieval of long COVID literature. We expect this system to improve search accuracy and reduce the time required for researchers to identify relevant studies, ultimately supporting faster scientific discovery and better understanding of long COVID.

## II. Prior work

Recent advancements in NLP and DL have significantly improved the ability to search, retrieve, and analyze large-scale biomedical literature, particularly in the context of COVID-19, and several search engine tools have been developed. Langnickel et al. introduced *preVIEW*, a search engine specifically designed for COVID-19 literature. This system leverages a transformer-based model, BERT, to classify research articles and distinguish long COVID-related studies from other COVID-19 publications. The model achieved a strong performance, with an F1-score of approximately 91%, demonstrating both high precision and recall [5]. This work highlights the effectiveness of transformer architectures in capturing semantic relationships within biomedical text. Furthermore, Esteva et al. developed an AI-powered search engine that combines BERT-based retrieval with keyword-based methods to identify relevant COVID-19 research. Their system goes beyond document retrieval by incorporating a question-answering framework, which generates concise summaries in response to user queries. This is achieved using transformer models alongside GPT-2, enabling more interactive and user-focused information extraction [9]. Another notable system is the AWS CORD-19 Search platform, which integrates neural search techniques with question answering (**QA**) and knowledge graph approaches. This system demonstrated superior performance in tasks such as document ranking, passage retrieval, and question answering over COVID-19 literature, showcasing the benefit of combining multiple AI methodologies within a unified framework [10]. Additionally, the COVID-19 Search Engine (CO-SE) employs a transformer-based architecture to extract precise answers from scientific literature. The system outperformed several baseline models, achieving approximately 71% Exact Match (**EM**) and 79% Semantic Answer Similarity (**SAS**), indicating its capability to both retrieve and semantically interpret relevant information from large datasets [11].

To achieve the goal of this project which is designing an effective search engine for classifying long COVID-related studies, we will build upon these advancements by leveraging deep learning techniques tailored for textual data. Transformer-based models (such as BERT) are particularly well-suited for this task due to their ability to capture contextual and semantic meaning in text. In addition, alternative sequential models such as Long Short-Term Memory (**LSTM**) networks may be explored for comparison. As a baseline, we will implement a Multi-Layer Perceptron (**MLP**) model using extracted textual features. The performance of more advanced models will then be evaluated against this baseline, as well as against existing systems such as preVIEW. This comparative approach will allow us to assess the effectiveness of different architectures in accurately classifying long COVID literature.

## III. Preliminary results

The dataset used in this project is obtained from Hugging Face [8], and contains a total of 552 COVID-19-related research articles. Each sample consists of the paper title and full abstract as input features. The corresponding labels indicate whether a paper is related to long COVID (label = 1) or not (label = 0). The dataset is split into 414 training samples and 138 test samples. As shown in Figure 1, the training set contains 215 long COVID papers (51.9%), while the test set includes 68 long COVID papers (49.3%). This indicates that the dataset is well balanced across

both classes in both splits, which reduces the risk of class imbalance bias during training and evaluation. Additionally, there are no missing labels in either the training or test sets, allowing all samples to be fully utilized without data exclusion.

As an initial baseline, a simple MLP model was implemented. The model consists of a single hidden layer with 64 neurons and uses the ReLU activation function. The training setup includes Cross-Entropy Loss, the Adam optimizer (learning rate = 0.001), batch size of 16, and training for 10 epochs. This baseline model achieved a test accuracy of 0.89 and an F1-score of 88.54%, demonstrating strong initial performance that is comparable to the preVIEW system reported in prior work [5]. However, given the simplicity of the model and dataset size, these results should be interpreted cautiously, as they may not fully reflect generalization performance on larger-scale datasets. A major concern is potential overfitting due to the small dataset size. While performance is strong on the current test split, the model may not generalize well to the broader COVID-19 literature data, which contains millions of publications. In real-world settings, only a small fraction of papers are related to long COVID, making generalization under class imbalance a critical challenge. To overcome these issues, we plan to incorporate regularization techniques such as dropout and weight decay, as well as explore more robust architectures like transformers and LSTMs.

This project is implemented in Python and leverages several machine learning and NLP libraries. Key tools include:

- datasets (Hugging Face) for loading the dataset
- matplotlib and seaborn for data visualization
- transformers (Hugging Face) for tokenization and transformer-based modeling
- AutoTokenizer (BERT-based) for preprocessing text inputs
- PyTorch for implementing MLP, LSTM, and deep learning models
- Numpy and pandas for data preprocessing and manipulation

For optimization, multiple optimizers such as Adam and Stochastic Gradient Descent (SGD) will be evaluated. In addition, hyperparameter tuning will be performed using a grid search strategy across:

- Batch sizes: 8, 16, 32
- Learning rates: 1e-5, 3e-5, 5e-5
- Hidden layers: 1, 3, 5, 10
- Epochs: 1 to 10

A significant bottleneck in this project is the computational cost associated with training transformer-based models, which require substantial memory and processing power. To address this limitation, lightweight baseline models such as MLPs are used for initial experiments. Next, we focus on implementing a transformer-based architecture for the proposed SENA (Search Engine with Neural Attention) system and evaluating its performance against baseline models. Future work will aim to enhance the SENA system to better handle large-scale and potentially imbalanced datasets.

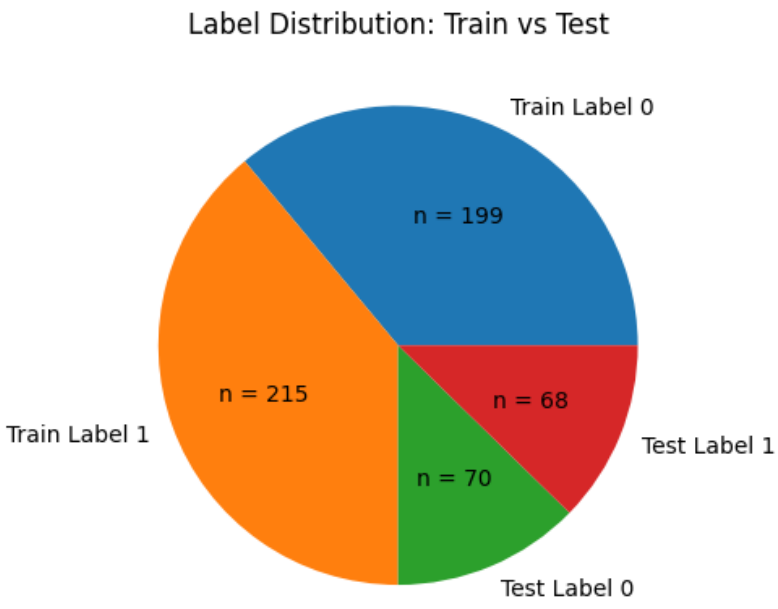


Figure 1. Distribution of long COVID and non-long COVID papers across the training and test splits.

## IV. Project Deliverables

Upon successful completion of this project, we aim to develop SENA as a search engine tool with neural attention, which is a competitive AI-based search and classification tool capable of distinguishing long COVID-related articles from other COVID-19 literature based only on their abstracts. A key benchmark for this project is the preVIEW system, which achieved an F1-score of approximately 91%. Our goal is to design and optimize SENA to achieve comparable performance, demonstrating that our approach can effectively replicate or improve results in this task. This project is structured around the following sub-goals:

1) **Model Optimization and Benchmarking**: We aim to systematically explore and optimize different deep learning architectures, including MLPs, LSTMs, and transformer-based models. The objective is to identify the model that yields the highest performance. Evaluation will be conducted using standard metrics, including accuracy, precision, recall, and F1-score, to ensure a comprehensive assessment of model effectiveness.
2) **Performance Evaluation Under Realistic Conditions**: Although the dataset used in this study is relatively balanced, real-world COVID-19 literature is highly imbalanced, with only a small fraction of publications focusing on long COVID. Therefore, a key sub-goal is to evaluate how well the model performs under conditions of class imbalance, with particular emphasis on F1-score.

3) **Generalization and Robustness Analysis**: As a future approach, we will assess the model's ability to generalize beyond the current dataset, ensuring that it does not overfit and can maintain strong performance when applied to larger and more diverse data.

All codes are available at: [https://github.com/Sina2285/stats507-coursework/tree/main/Final-Project](https://github.com/Sina2285/stats507-coursework/tree/main/Final-Project)

## V. Timeline

| | Week 1 | Week 2 | Week 3 | Week 4 | Week 5 |
|---|---|---|---|---|---|
| Data Selection | | | | | |
| Literature Review | | | | | |
| Proposal Writing | | | | | |
| Preliminary Results | | | | | |
| Sub-aim 1 | | | | | |
| Sub-aim 2 | | | | | |
| Final Project Report | | | | | |

## VI. References


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